KICKOFF MEETING

Building Our Antweight Combat Robot

One pound of pure attitude, controlled from your phone

WEIGHT CLAS
Antweight · 1 lb

CONTRO
Wi-Fi + Browser

TODAY'S SESSION

Agenda

01 What is antweight combat robotics

02 Meet our robot, AntBot-S3

03 What we'll need to build it

04 How we'll build it, step by step

05 Driving and safety basics

06 Getting started together

THE SPORT

What Is Antweight Combat Robotics

Two remote-controlled robots, one arena, three minutes. Builders design and drive machines armed with spinners, wedges, and hammers — and the last one still moving wins.

1 lb
Antweight class limit

3 min
Typical match length

THE SPORT

How A Match Is Won

FASTEST WIN

Knockout

Immobilize the other bot completely and the match ends early — the clearest win in the sport.

IF TIME RUNS OUT

Judges' Decision

Judges score on damage dealt, control, and aggression to pick a winner.

MANY FORMS

Weapon Styles Vary

Spinners, flippers, wedges, and hammers are all common — our build uses a spinning weapon.

OUR ROBOT

Meet AntBot-S3

A microcontroller board is the entire brain of our robot — it makes its own Wi-Fi network, so a phone or laptop can drive it straight from a browser. No transmitter, no receiver, no hobby radio gear.

BRAIN

ESP32-S3 DevKit

Hosts the Wi-Fi network and web page, and turns joystick input into motor commands.

DRIVE

DRV8833 + 2× N20

Dual motor driver spins two gear motors for tank-style steering, and brakes to a stop.

WEAPON

Brushless ESC + Motor

A slim speed controller spins the weapon motor — throttle only, no reverse.

OUR ROBOT

Driven By Wi-Fi, Not Radio

1. The robot creates its own Wi-Fi network on power-up.
2. Your phone or laptop joins that network and opens a browser.
3. A joystick and weapon slider load right on the page — no app to install.

on-screen joystick, phone browser

THE BUILD

What We'll Need

PART	SPEC	QTY
2S LiPo Battery	7.4V nominal, high discharge	1
ESP32-S3 DevKit	Board with onboard RGB status LED	1
DRV8833 Motor Driver	2-channel H-bridge breakout	1
N20 Gear Motors	~500 RPM class @ 6V	2
Brushless Weapon ESC	Slim profile, 15A, 2S, throttle-only	1
Weapon Motor	Brushless, sized to the ESC	1
Connectors & Wire	JST-XH, XT30, silicone wire, heat-shrink	—

THE BUILD

How We'll Build It

01
Set Up Software
Install the Arduino IDE and get the robot's code onto your computer.

02
Plan The Layout
Dry-fit battery, driver, brain, and ESC in the shell before cutting wire.

03
Wire The Harness
Build power, signal, and motor connections following the pinout.

04
Flash Firmware
Set the pin numbers and Wi-Fi name, then upload to the board.

05
Test & Drive
Run the checklist, then drive it lifted off the ground first.

THE BUILD

Wiring It Up

Every signal on the board comes straight from its own GPIO pins — five wires tell the robot how to move and fire.

SIGNAL	GPIO	GOES TO
WEAPON_PIN	6	Weapon ESC signal
L_IN1 / L_IN2	7 / 8	Left motor driver
R_IN2 / R_IN1	9 / 10	Right motor driver

KEEP IT SIMPLE

One shared power bus, short direct signal runs, and keyed connectors — nothing can plug in backward.

ONLY TWO PLUGS

Everything else solders direct — the battery connector and the weapon signal pigtail are the only joints you'll unplug.

DRIVING & SAFETY

Every Boot Starts Stopped

Boot

Motors braked, weapon held at zero throttle for 3 seconds while the ESC arms.

Ready

LED turns solid cyan — Wi-Fi is up, waiting for a browser to connect.

Armed

Tap ACTIVATE in the browser — LED turns green, bot is now live at zero throttle.

Failsafe

Lose the link or hit STOP ALL for 1.2 seconds and it re-latches instantly.

DRIVING & SAFETY

Driving The Bot

1

Join the robot's Wi-Fi network from your phone or laptop.

2

Open a browser to the robot's address — the joystick page loads automatically.

3

Tap ACTIVATE, then drag the pad to drive and the slider to spin the weapon.

4

Tap STOP ALL — or hit Escape — any time to hard-stop instantly.

DRIVING & SAFETY

Before Every Run

- [ ] Battery voltage checked before connecting.
- [ ] Weapon area clear of fingers and tools.
- [ ] First test drive done with wheels off the ground.
- [ ] Failsafe confirmed by closing the browser tab.
- [ ] No exposed wiring, all heat-shrink intact.
- [ ] Status LED goes solid cyan before activating.
- [ ] Weapon tested briefly at low throttle first.
- [ ] STOP ALL tested and re-latch confirmed.

DRIVING & SAFETY

Rules We Never Skip

- Disconnect the main battery whenever the bot isn't actively being driven.
- Keep hands, tools, and loose clothing clear of the weapon any time the battery is connected.
- Wear eye protection and restrain the robot during weapon testing.
- Re-run the full pre-flight checklist after any rewiring or firmware change.

LET'S GET STARTED

From Parts List To Arena Ready

One board, one battery, and a browser — everything else is just following the steps together.

- See you at the build table.